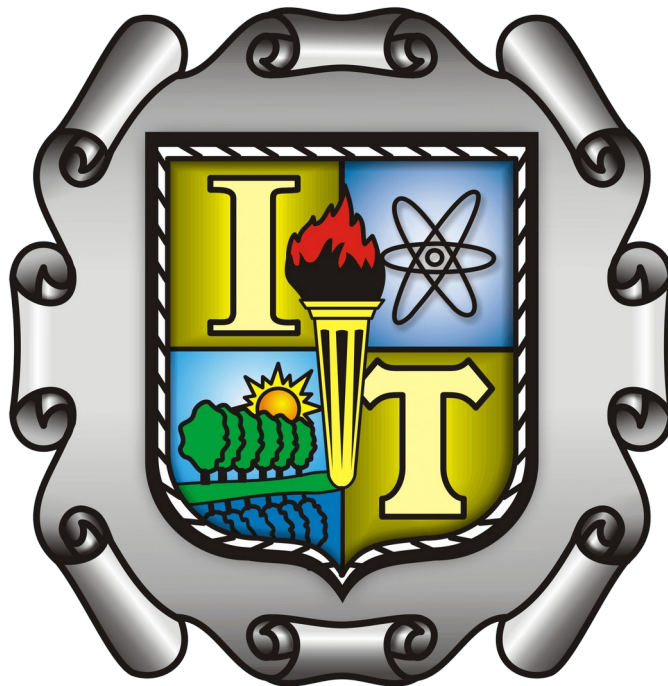


Tecnologico nacional de mexico

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Análisis de tráfico de red con Wireshark para la prevención de ataques



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20051204

Edición Visualización Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

Wlp0s20f3

http

Preferencias de botones de filtro...

Etiqueta:

Filtro:

Aceptar Cancelar

Comentario:

Aceptar Cancelar

No.	Time	Source	Destination	Protocol	Length	Info
950	27.8352024880	2009:2:0:3208:fcd5:5...	2009:2:0:3208:fcd5:5...	HTTP	463	GET / HTTP/1.1
962	27.981559189	2006:4780::6812:1a78	2006:4780::6812:1a78	HTTP	91	HTTP/1.1 200 OK (text/html)
991	28.187957217	2006:2f0:3280:fcd5:5...	2006:4780::6812:1a78	HTTP	452	GET /favicon.ico HTTP/1.1
994	28.135439642	2006:4780::6812:1a78	2006:2f0:3280:fcd5:5...	HTTP	91	HTTP/1.1 404 Not Found (text/html)
1027	37.372834792	2006:2f0:3280:fcd5:5...	2006:9900:27f2:400:1...	HTTP	159	GET / HTTP/1.1
1029	37.395671542	2006:9900:27f2:400:1...	2006:2f0:3280:fcd5:5...	HTTP	447	HTTP/1.1 204 No Content
1051	41.379585376	192.168.100.7	3.167.246.11	HTTP	138	GET / HTTP/1.1
1053	41.400328311	3.167.246.11	192.168.100.7	HTTP	427	HTTP/1.1 204 No Content

Frame 950: 438 bytes on wire (3504 bits), 438 bytes captured (3504 bits) on interface wlp0s20f3, Ethernet II, Src: Intel 802c:f5 (0:77:c3:06:2c:f5), Dst: HuaweiTechno_4a:51:c0 (24:91:bb:4a:51:c0)

Internet Protocol Version 6, Src: 2006:2f0:3280:fcd5:5d86:e15f:d58e:d9e2, Dst: 2006:4780::6812:1a78

Transmission Control Protocol, Src Port: 38202, Dst Port: 80, Seq: 1, Ack: 1, Len: 352

Hypertext Transfer Protocol

<pre> 0000 24 91 bb 4a 51 c0 f0 77 c3 06 2c f5 86 dd 60 ef 0010 08 0d 01 00 06 40 28 06 02 f0 32 00 fc dd 5d 86 0020 e1 5f d5 8e d9 e2 26 06 47 00 00 00 00 00 00 0030 00 00 68 12 1a 78 95 3a 00 50 f3 64 1e 5f 0d 00 0040 12 2e 80 18 00 49 39 c3 00 00 01 01 08 0a 75 1b 0050 a4 9c fc 0b 35 e5 47 45 64 20 2f 20 48 54 54 58 0060 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 65 78 61 6d 0070 70 6c 65 2e 63 6f 6d 0d aa 55 73 65 72 2d 41 67 0080 65 6e 74 3a 20 4d 6f 7a 69 6c 6e 61 2f 35 2e 30 0090 20 28 58 31 31 3b 20 4c 69 6e 75 78 20 78 38 36 00a0 5f 36 34 3b 20 72 76 3a 31 34 38 2e 30 29 20 47 00b0 65 63 6b 6f 2f 32 30 31 30 39 31 38 31 20 46 69 00c0 72 65 66 6f 78 2f 31 34 2e 38 6d 0a 41 63 63 00d0 65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61 00e0 70 70 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 00f0 2b 78 6d 6c 2c 61 70 70 69 63 61 74 69 6f 6e 0100 2f 78 6d 6c 3b 71 3d 30 2e 39 2c 2a 2f 2a 3b 71 0110 30 39 2e 38 0d 0a 41 63 65 70 74 2d 4c 61 6e 0120 67 75 61 6f 65 3a 20 65 73 2d 45 53 2c 65 73 3b 0130 71 3d 30 2e 39 2c 65 6e 2d 65 53 3b 71 3d 30 2e 0140 38 2c 65 6e 3b 71 3d 30 2e 37 0d 0a 41 63 63 65 0150 70 74 2d 45 6e 63 6f 64 69 6e 74 3a 20 6f 7a 69 </pre>	<pre> \$-JQ;w m...@ ...& ...x: ...:GE T / /1.1:Ho st: ple.com ent: Moz illa/ (XII; L inux _64; rv: ecko/201 00101 refox/10 epr: tex t/ pplicati on/ +xml,app /xml;q=0.9,/ e=0.9;Ac cept= guage: e s- q=0.9,en- 8,en;q=0 pt-Enco ding: </pre>
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100%

Paquetes: 1123 - Mostrado: 8 (0.7%) - Perdido: 0 (0.0%)

Perfil: Defalta

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Preferencias de botones de filtro... Etiqueta: [Introduzca una descripción para el botón de filtro] Filtro: **http**

Comentario: [Introduzca un comentario para el botón de filtro] Aceptar Cancelar

No.	Time	Source	Destination	Protocol	Length	Info
937	27.810865801	fe80::b669:667d:d6a2...	fe80::1	DNS	182	Standard query 0xfaf5 HTTPS example.com OPT
938	27.818356148	fe80::b669:667d:d6a2...	fe80::1	DNS	182	Standard query 0x7dff A example.com OPT
939	27.818681388	fe80::b669:667d:d6a2...	fe80::1	DNS	182	Standard query 0xf9d9 AAAA example.com OPT
940	27.821678826	fe80::1	fe80::b669:667d:d6a2...	DNS	134	Standard query response 0x7dff A example.com A 184.18.26.128 A 184.18.27.128 OPT
941	27.823878197	fe80::1	fe80::b669:667d:d6a2...	DNS	260	Standard query response 0xfaf5 HTTPS example.com HTTPS A 184.18.27.128 A 184.18.26.128 AAAA 2686:4780::6812...
942	27.824251172	fe80::1	fe80::b669:667d:d6a2...	DNS	153	Standard query response 0xb929 A incoming.telemetry.mozilla.org OPT
955	27.934519361	fe80::b669:667d:d6a2...	fe80::1	DNS	121	Standard query 0xb929 A incoming.telemetry.mozilla.org OPT
966	27.934643850	fe80::b669:667d:d6a2...	fe80::1	DNS	121	Standard query 0xc64 AAAA incoming.telemetry.mozilla.org OPT
971	27.943234553	fe80::1	fe80::b669:667d:d6a2...	DNS	196	Standard query response 0xb929 A incoming.telemetry.mozilla.org CNAME telemetry-incoming.r53-2.services.moz...
972	27.945588901	fe80::1	fe80::b669:667d:d6a2...	DNS	259	Standard query response 0xc64 AAAA incoming.telemetry.mozilla.org CNAME telemetry-incoming.r53-2.services.moz...
975	27.946229191	fe80::b669:667d:d6a2...	fe80::1	DNS	136	Standard query 0xb908 AAAA telemetry-incoming.r53-2.services.mozilla.org OPT
976	27.955287222	fe80::1	fe80::b669:667d:d6a2...	DNS	218	Standard query response 0xc64 AAAA telemetry-incoming.r53-2.services.mozilla.org SOA ns-1507.awsdns-60.org...
977	28.022265564	fe80::b669:667d:d6a2...	fe80::1	DNS	99	Standard query 0xc11 HTTPS iana.org OPT
978	28.022342365	fe80::b669:667d:d6a2...	fe80::1	DNS	99	Standard query 0x5ab3 A iana.org OPT
979	28.022401236	fe80::b669:667d:d6a2...	fe80::1	DNS	99	Standard query 0xc452 AAAA iana.org OPT
987	28.061950150	fe80::1	fe80::b669:667d:d6a2...	DNS	127	Standard query response 0xc452 AAAA iana.org AAAA 2001:509:88:200::8 OPT
988	28.061958171	fe80::1	fe80::b669:667d:d6a2...	DNS	115	Standard query response 0x5ab3 A iana.org A 192.0.43.8 OPT
990	28.082468910	fe80::1	fe80::b669:667d:d6a2...	DNS	153	Standard query response 0xc11 HTTPS iana.org SOA sns.dns.icann.org OPT

Frame 942: 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface wlan2ef03,
 Ethernet II, Src: HuaweiTechno_Aa:51:c8 (24:91:bb:a4:51:c8), Dst: Intel_E6:2c:f5 (f8:77:c3:06:2c:
 Internet Protocol Version 6, Src: fe80::1, Dst: fe80::b669:667d:d6a2:429d
 User Datagram Protocol, Src Port: 53, Dst Port: 33576
 Domain Name System (response)

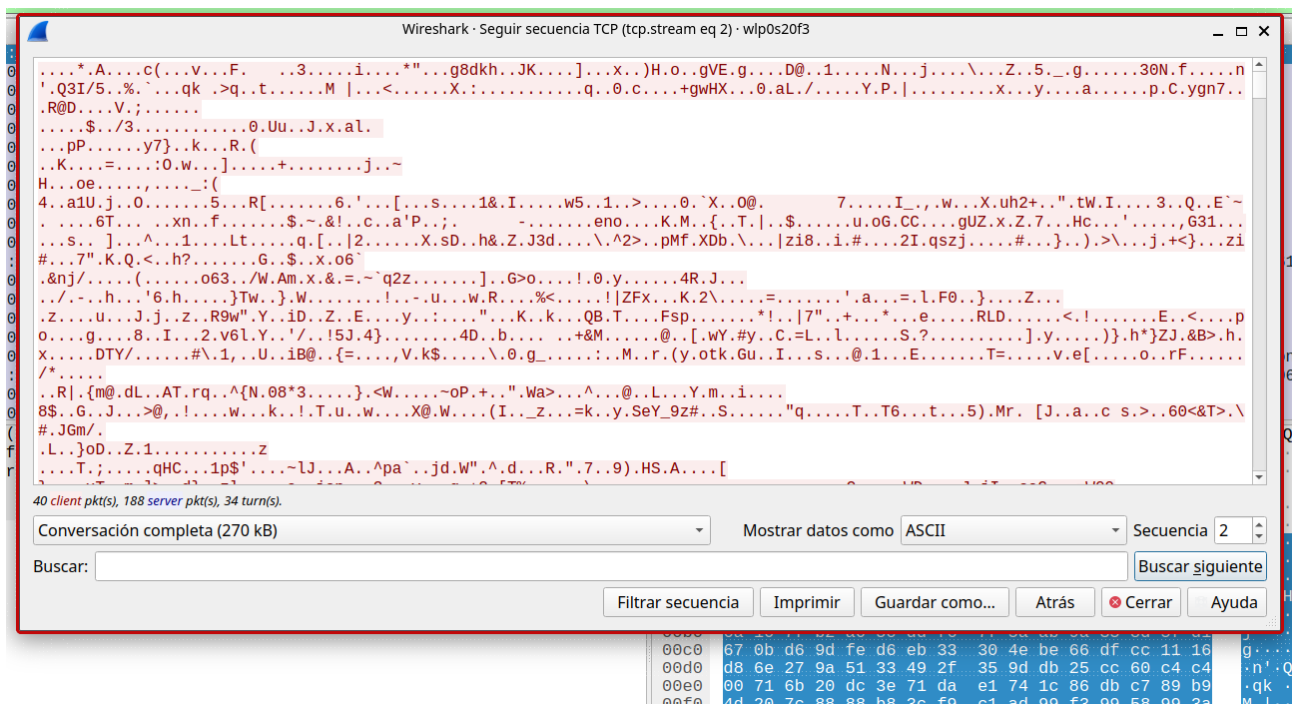
Packets: 1123 - Mostrado: 100 (8.9%) - Perdido: 0 (0.0%) Perfil: Default

captura del TCP

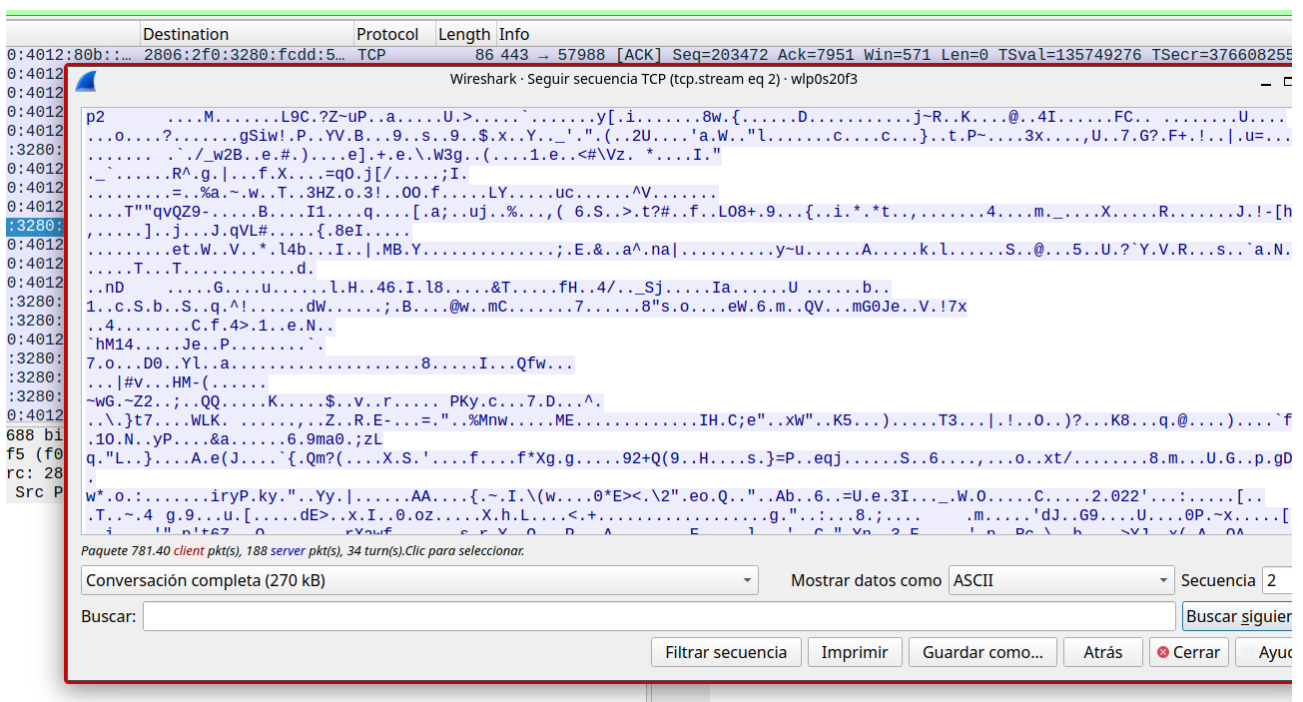
Wireshark packet capture showing a TCP stream. The packet list on the left shows a sequence of packets, with packet 162 selected. The packet details pane on the right shows the structure of the selected packet, including the Ethernet II header, Internet Protocol Version 6 header, and Transmission Control Protocol header. The packet bytes pane at the bottom shows the raw data of the selected packet.

seguir un http por tcp stream

Wireshark packet capture showing a TCP stream. The packet list on the left shows a sequence of packets, with packet 162 selected. The packet details pane on the right shows the structure of the selected packet, including the Ethernet II header, Internet Protocol Version 6 header, and Transmission Control Protocol header. The packet bytes pane at the bottom shows the raw data of the selected packet.



las demas estan cifradas por https



¿Qué información es visible?

En el tráfico HTTP se puede observar:

- URLs completas
- Métodos GET
- Información en texto plano
- Datos enviados entre cliente y servidor

En el tráfico DNS:

- Los dominios consultados
- Las direcciones IP asociadas

¿Qué riesgo existe?

El riesgo principal es que un atacante puede:

- Interceptar información sensible
- Robar credenciales
- Espiar la actividad del usuario
- Realizar ataques Man-in-the-Middle

Diferencia entre HTTP y HTTPS

HTTP:

- No tiene cifrado
- Toda la información es visible
- Es inseguro

HTTPS:

- La información está cifrada
- No se pueden ver los datos
- Es seguro

Conclusión

La práctica demuestra que el tráfico de red puede ser interceptado fácilmente si no está protegido. El uso de HTTPS es fundamental para garantizar la seguridad de la información y evitar ataques.